

SQL Window Functions — Practice Questions

Table Setup

```
CREATE DATABASE IF NOT EXISTS window_practice;
USE window_practice;
```

```
CREATE TABLE marks (
    student_id INTEGER PRIMARY KEY AUTO_INCREMENT,
    name        VARCHAR(100) NOT NULL,
    branch      VARCHAR(10)  NOT NULL,
    marks       INTEGER NOT NULL
);
```

```
INSERT INTO marks (name, branch, marks) VALUES
('Nitish', 'EEE', 82), ('Rishabh', 'EEE', 91),
('Anukant', 'EEE', 69), ('Rupesh', 'EEE', 55),
('Shubham', 'CSE', 78), ('Ved', 'CSE', 43),
('Deepak', 'CSE', 98), ('Arjun', 'CSE', 95),
('Vinay', 'ECE', 95), ('Ankit', 'ECE', 88),
('Rohit', 'ECE', 95), ('Anand', 'ECE', 81),
('Prashant', 'MECH', 75), ('Sunny', 'MECH', 39);
```

```
CREATE TABLE monthly_sales (
    id      INTEGER PRIMARY KEY AUTO_INCREMENT,
    month   INTEGER NOT NULL,
    year    INTEGER NOT NULL,
    sales   DECIMAL(10,2) NOT NULL
);
```

```
INSERT INTO monthly_sales (month, year, sales) VALUES
(1, 2020, 100), (2, 2020, 150), (3, 2020, 200), (4, 2020, 175),
(5, 2020, 250), (6, 2020, 300), (7, 2020, 275), (8, 2020, 200);
```

Topic 1: ROW_NUMBER / RANK / DENSE_RANK

1. Assign a row number to each student ordered by marks descending (global).
2. Rank all students by marks descending using RANK() — observe how ties are handled.
3. Rank all students by marks descending using DENSE_RANK() — compare with RANK().
4. Show all three: ROW_NUMBER, RANK, DENSE_RANK together for all students ordered by marks descending.
5. Rank students within each branch by marks descending using RANK().
6. Rank students within each branch by marks descending using DENSE_RANK().
7. Generate roll numbers in the format "BRANCH-N" for each student (e.g., CSE-1, ECE-2).

Topic 2: PARTITION BY with Aggregates

8. Show each student's marks alongside the average marks of their branch (without GROUP BY).
9. Find students who scored below the average marks of their own branch.
10. Show each student's marks and the maximum marks in their branch.
11. Rank customers within each month by their total spending (Zomato orders dataset).
12. Rank batsmen within each IPL team by total runs using DENSE_RANK.
13. Find top 5 batsmen per IPL team (rank $\leq$ 5).

Topic 3: FIRST_VALUE / LAST_VALUE / NTH_VALUE

14. Show the topper's name (highest marks) alongside every student row (global).
15. Show the branch topper's name alongside every student row (per branch).
16. Find the last (lowest marks) student's name in each branch — use FRAME clause.
17. Find the 2nd highest marks student in each branch using NTH_VALUE.
18. Find the 5th topper of each branch using NTH_VALUE.
19. Find the branch toppers — show only their actual rows (not all rows with topper name repeated).
20. Find the bottom students of each branch (their actual rows only) using LAST_VALUE + FROM subquery.

Topic 4: Frames

21. What is the default frame when you write `ORDER BY col` inside OVER() without specifying ROWS BETWEEN?
22. Why does LAST_VALUE without a FRAME clause not return the actual last value of the partition? How do you fix it?
23. Write a query using `ROWS BETWEEN 1 PRECEDING AND 1 FOLLOWING` — what rows does it include?
24. Write a query that includes ALL rows in the partition for the frame.

Topic 5: WINDOW Alias

25. Rewrite Q16 (LAST_VALUE with frame) using a WINDOW alias `w` to avoid repeating the full window spec.
26. Rewrite Q19 (branch toppers) using WINDOW alias for FIRST_VALUE.

Topic 6: LEAD & LAG

- 27. Show each student's marks alongside the previous student's marks (by student_id).
- 28. Find the Month-over-Month revenue growth percentage for Zomato (orders dataset).
- 29. Show each student's marks and the next student's marks (using LEAD).

Topic 7: Cumulative Sum / Average

- 30. Calculate the cumulative sales total month by month (monthly_sales table).
- 31. Calculate V Kohli's career runs and career average after each match (IPL dataset).
- 32. Calculate the cumulative average score for each student across their tests.

Topic 8: Running Average / Percent of Total / Percent Change

- 33. Calculate a 3-month moving average for monthly sales.
- 34. Find what percentage of total votes each movie holds (movies dataset).
- 35. Calculate the month-over-month percentage change in Zomato revenue.

Bonus / Mixed

- 36. Find the top 3 most run-scoring batsmen per IPL team — show their details.
- 37. For each branch, find students who scored within 10 marks of the branch topper.
- 38. Show running total of marks for students ordered by student_id within each branch.